

PRACTICAL SESSION 1

TASK 1

```
[(base) saiabburi@Geyasree-MacBook-Air ~ % mkdir Session1
[(base) saiabburi@Geyasree-MacBook-Air ~ % cd Session1
[(base) saiabburi@Geyasree-MacBook-Air Session1 % pwd
/Users/saiabburi/Session1
[(base) saiabburi@Geyasree-MacBook-Air Session1 % nano process.c
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild Process Started\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID from Child: %d\n", getppid());

        execlp("ls", "ls", "-l", NULL);

        // This executes only if execlp fails
        perror("exec failed");
        exit(1);
    }
    else {
        // Parent process
        printf("\nParent is waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
        printf("Parent process completed.\n");
    }

    return 0;
}
```

```

(base) saiabburi@Geyasree-MacBook-Air Session1 % ls
process.c
(base) saiabburi@Geyasree-MacBook-Air Session1 % clang process.c -o process
(base) saiabburi@Geyasree-MacBook-Air Session1 % ls
process
process.c
(base) saiabburi@Geyasree-MacBook-Air Session1 % ./process
Parent Process Started
Parent PID: 8922

Parent is waiting for child...

Child Process Started
Child PID: 8932
Parent PID from Child: 8922
total 80
-rwxr-xr-x  1 saiabburi  staff   33716 Aug 21 21:06 process
-rw-r--r--  1 saiabburi  staff    877 Aug 21 21:05 process.c
Child process completed.
Parent process completed.

```

TASK 2

```

(base) saiabburi@Geyasree-MacBook-Air Session1 % lscpu
zsh: command not found: lscpu
(base) saiabburi@Geyasree-MacBook-Air Session1 % lsblk
zsh: command not found: lsblk
(base) saiabburi@Geyasree-MacBook-Air Session1 % uname -a
Darwin Geyasree-MacBook-Air.local 22.5.0 Darwin Kernel Version 22.5.0: Thu Jun  8 22:21:34 PDT 2023; root:xnu-8796.121.3~7/RELEASE_ARM64_T8112 arm64
(base) saiabburi@Geyasree-MacBook-Air Session1 % sw_vers
ProductName:      macOS
ProductVersion:   13.4.1
ProductVersionExtra: (c)
BuildVersion:     22F770820d

```

```

(base) saiabburi@Geyasree-MacBook-Air Session1 % sysctl -n hw.ncpu
8
(base) saiabburi@Geyasree-MacBook-Air Session1 % free -h
zsh: command not found: free
(base) saiabburi@Geyasree-MacBook-Air Session1 % df -h

```

Filesystem	Size	Used	Avail	Capacity	iuused	ifree	%iused	Mounted on
/dev/disk3s3s1	228Gi	11Gi	2.4Gi	83%	355382	24689480	1%	/
devfs	201Ki	201Ki	0Bi	100%	694	0	100%	/dev
/dev/disk3s6	228Gi	2.0Gi	2.4Gi	47%	2	24689480	0%	/System/Volumes/VM
/dev/disk3s4	228Gi	9.2Gi	2.4Gi	80%	1195	24689480	0%	/System/Volumes/Preboot
/dev/disk3s2	228Gi	721Mi	2.4Gi	24%	299	24689480	0%	/System/Volumes/Update
/dev/disk1s2	500Mi	6.0Mi	480Mi	2%	1	4913840	0%	/System/Volumes/xarts
/dev/disk1s1	500Mi	6.3Mi	480Mi	2%	41	4913840	0%	/System/Volumes/iSCPreboot
/dev/disk1s3	500Mi	3.1Mi	480Mi	1%	63	4913840	0%	/System/Volumes/Hardware
/dev/disk3s1	228Gi	201Gi	2.4Gi	99%	1350700	24689480	5%	/System/Volumes/Data
map auto_home	0Bi	0Bi	0Bi	100%	0	0	100%	/System/Volumes/Data/home
/dev/disk3s3	228Gi	11Gi	2.4Gi	83%	356884	24689480	1%	/System/Volumes/Update/mnt1
/Applications/aha.app/Wrapper	228Gi	202Gi	2.8Gi	99%	1354759	28909840	4%	/private/var/folders/kt/010117zs6wv1hzwk1chttnmw0000gn/X/B83E0D7F-A62E-5E2C-8EF2-D2D2C11857F6
/Applications/Savana.app/Wrapper	228Gi	203Gi	1.7Gi	100%	1351682	17419320	7%	/private/var/folders/kt/010117zs6wv1hzwk1chttnmw0000gn/X/451891B5-1EA6-5012-9613-1487228FAE00

```

(base) saiabburi@Geyasree-MacBook-Air Session1 % lsusb
zsh: command not found: lsusb
(base) saiabburi@Geyasree-MacBook-Air Session1 % lscpu
zsh: command not found: lscpu
(base) saiabburi@Geyasree-MacBook-Air Session1 % top

```

PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORTS	MEM	PURG	CMPRS	PGRP	PPID	STATE	BOOSTS
368	WindowServer	12.4	89:37.73	18	5	2053	411M-	17M+	110M	368	1	sleeping	*0[1]
0	kernel_task	6.1	47:05.25	456/8	0	0	15M	0B	0B	0	0	running	0[0]
8995	top	5.5	00:00.93	1/1	0	33	5809K+	0B	0B	8995	8833	running	*0[1]
8996	screencaptur	1.8	00:00.54	2	1	66	8146K	752K	0B	591	591	sleeping	*0[488+]
8614	Google Chrom	1.5	01:04.49	46	1	910	271M+	0B	122M-	8614	1	sleeping	*0[186]
8830	Terminal	0.7	00:11.43	8	3	257	59M	26M	8384K	8830	1	sleeping	*0[169]
734	Grammarly De	0.5	03:29.20	12	1	320	61M	0B	32M	734	1	sleeping	*0[3285]
883	app_inkwell	0.4	04:16.45	16	1	342	32M	0B	16M	883	734	sleeping	*0[2619]
8625	Google Chrom	0.4	00:11.02	17	1	164	54M+	0B	24M-	8614	8614	sleeping	*0[3]
8720	Google Chrom	0.4	00:12.31	19	1	181-	134M-	0B	71M-	8614	8614	sleeping	*0[10]
8728	Google Chrom	0.3	00:29.73	35	1	425	465M+	0B	134M	8614	8614	sleeping	0[5]
8997	screencaptur	0.2	00:00.10	4	2	147	7649K	0B	0B	8997	1	sleeping	*0[308+]
392	airportd	0.2	03:43.05	8	6	556	18M	0B	11M	392	1	sleeping	*872[1]
669	mysqld	0.2	01:48.29	38	0	63	391M	0B	388M	669	1	sleeping	*0[1]
360	bluetoothd	0.2	04:41.88	9	3	334+	9489K+	0B	4448K	360	1	sleeping	*0[1]
304	fseventsd	0.2	02:49.77	10	1	146	5345K	0B	2000K	304	1	sleeping	*0[1]
8699	Google Chrom	0.1	00:39.00	30	1	371	240M+	0B	92M	8614	8614	sleeping	0[8]
6896	fileprovider	0.1	00:12.90	6	5	129	12M	0B	8320K	6896	1	sleeping	0[269]
549	distnoted	0.1	00:16.53	2	1	307	3057K	0B	864K	549	1	sleeping	*0[1]
888	com.apple.We	0.1	02:29.65	4	1	101	21M-	48K	11M	888	1	sleeping	*0[1664]
363	corebrightne	0.1	01:42.88	3	2	128	3393K	0B	1360K	363	1	sleeping	*0[1]
523	searchpartyd	0.1	00:02.53	4	3	85+	6465K+	0B	3424K	523	1	sleeping	*0[1434+]
351	distnoted	0.1	00:12.01	2	1	167	1777K	0B	624K	351	1	sleeping	*0[1]
345	locationd	0.1	00:45.84	5	2	242+	10M+	0B	6064K-	345	1	sleeping	*0[3269+]
312	powerd	0.1	00:22.75	3	2	133	4289K	0B	1600K	312	1	sleeping	*0[1]
922	com.apple.We	0.1	01:36.59	4	1	78	187M	0B	159M	922	1	sleeping	*0[38835]
635	WirelessRadi	0.0	00:07.28	3	2	62+	3201K+	0B	1536K	635	1	sleeping	*0[1]
5363	Google Drive	0.0	00:23.05	128	1	575	68M	16K	34M	666	666	sleeping	*0[235]
685	sharingd	0.0	00:31.01	3	1	306	13M	0B	8864K	685	1	sleeping	*0[1]
670	com.muse.aut	0.0	00:24.25	52	3	34182+	4913K	0B	3168K	670	1	sleeping	*0[1]
592	Finder	0.0	00:48.19	5	2	479	110M	0B	74M	592	1	sleeping	*0[4178]
474	audioclocksy	0.0	00:18.96	2	1	64	6177K	0B	5040K	474	1	sleeping	*0[1]
7017	PerfPowerSer	0.0	00:29.62	5	3	315	12M	128K	3152K	7017	1	sleeping	0[243]